Last Name: First Name:

Matriculation Number:

Exam
Artificial Intelligence 2

April 2, 2026

Please ignore the QR codes; do not write on them, they are for grading support

	To be used for grading, do not write here											
prob.	1.1	1.2	2.1	2.2	3.1	3.2	4.1	4.2	4.3	5.1	Sum	grade
total	8	6	8	9	12	10	9	9	8	11	90	
reached												

Old exams are great practice resources, but since exam styles evolve, recent exams are better models, and any course topic even if not asked before can still appear in upcoming exams. Note that these exams are for learning support only, are not exhaustive or guaranteed correct, and may not reflect the actual quiz or exam questions.

The exams here are not necessarily identical to the old exams. For technical reasons, the points may change and some sub-problems that were elided in teh exam may appear here.

In the courses Artificial Intelligence I/II and SMAI we award bonus points for the first student who reports a factual error in an old exam. (Please report spelling/formatting errors as well.)

1 Probabilities

Problem 1.1 (Python)
Assume that X and Y are random variables with domains $\{0,\ldots,m\}$ and $\{0,\ldots,n\}$, respectively. In the Python programs below, the argument C holds the *conditional* probability distribution of Y given X in such a way that $C[x][y] = P(Y = y \mid X = x)$.

1. Which probability-related operation does the function `foo` compute? 4 Points

```
def foo(C,PX):
  res = []
  for y in range(len(C[0])):
    py = 0
    for x in range(len(C)):
      py += PX[x]*C[x][y]
    res.push(py)
  return res
```

Solution: It returns the probability distribution of Y if the probability distribution of X is given by `PX`.

2. In the code below, `bar(C,y)` is meant to return the probability $P(X + Y = s)$, where C is as above. Explain why it is buggy and indicate how it could be fixed. 4 Points

```
def bar(C,s):
  r = 0
  for x in range(len(C)):
    for y in range(len(C[x]):
      if x+y==s:
        r += C[x][y]
  return r
```

Solution: The function would be correct if C held the *joint* probability distribution, not the conditional one. Without the former, we do not have enough data to compute $P(X+Y=s)$. We would need to change the inputs to have more/different data.

Problem 1.2 (Working with Distributions)
Assume Boolean random variables X, Y, Z. Their joint probability distribution is given as follows:

x	y	z	$P(X=x, Y=y, Z=z)$
0	0	0	a
0	0	1	b
0	1	0	c
0	1	1	d
1	0	0	e
1	0	1	f
1	1	0	g
1	1	1	h

1. Assume the sample space Ω is $\{0,1\}^3$. How do we calculate the value of X as a function of a sample $e \in \Omega$? 1 Points

Solution: $X = e_1$, where e_1 is the first component of the triple e.

2. How many out of the values $a, \ldots, h$ can be dropped without losing information? 1 Points

Solution: 1 value (It does not matter which one because all values sum to 1.)

3. In terms of $a, \ldots, h$, give $P(X = Y \mid Z = 0)$. 2 Points

Solution: $(a + g)/(a + c + e + g)$

4. Now assume $e = f = g = h = 0$. Which calculations do you have to perform on a, b, c, d to check if Y and Z are independent? 2 Points

Solution: Put $p = a + b$ and $q = a + c$. Then check $a = pq$, $b = p(1 - q)$, $c = (1 - p)q$, and $d = (1 - p)(1 - q)$ (any one of which can be dropped).
There are other, equivalent sets of equations that are correct solutions. For example, we can check (three out of) $a = (a+b)(a+c)$, $b = (a+b)(b+d)$, $c = (c+d)(a+c)$, and $d = (c+d)(b+d)$.

2 Bayesian Reasoning

Problem 2.1 (Bayes' Rule)
Assume you are trying to predict whether a particular topic comes up in an exam. You have collected the following data:

- 30% of all topics come up in the exam.
- 40% of all topics come up in the tutorials.
- If a topic comes up in an exam, it was covered by an assignment 60% of the time.
- If a topic comes up in an exam, it came up in a tutorial 80% of the time.

You model this situation using 3 Boolean random variables X (comes up in exam), S (covered by assignments), and T (came up in a tutorial).

1. By filling in the gaps below, state for each number in the text above, which probability it describes. 2 Points
   1. $P($ `X=true` $) = 0.3$
   2. $P($ `T=true` $) = 0.4$
   3. $P($ `S=true` $\mid$ `X=true` $) = 0.6$
   4. $P($ `T=true` $\mid$ `X=true` $) = 0.8$

2. Assume the topic you are interested in **did not** come up in the exam. Argue briefly (sketch an argument) if and how we can obtain the probability that it was covered by a tutorial. 3 Points

Solution: We write t for $T = true$ and accordingly for x. The joint distribution of X and T contains 4 values: $P(t,x)$, $P(t,\neg x)$, $P(\neg t,x)$ and $P(\neg t,\neg x)$.
We know 4 properties about them:

- $P(x) = P(t,x) + P(\neg t,x) = 0.3$
- $P(t) = P(t,x) + P(t,\neg x) = 0.4$
- $P(t \mid x) = P(t,x)/P(x) = 0.8$
- $P(t,x) + (t,\neg x) + P(\neg t,x) + P(\neg t,\neg x) = 1$

From those we can compute the joint distribution as

	t	$\neg t$	Σ
x	0.24	0.06	0.3
$\neg x$	0.16	0.54	0.7
Σ	0.4	0.6	1

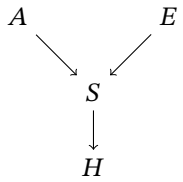
And from that, we can compute $P(t \mid \neg x) = P(t,\neg x)/P(\neg x) = 0.16/0.7 = 8/35 \approx 0.229$.
A direct calculation can be done as follows: $P(t \mid \neg x) = P(t,\neg x)/P(\neg x)$
$= \big(P(t) - P(t,x)\big)/(1 - P(x))$
$= \big(P(t) - P(t \mid x) \cdot P(x)\big)/(1 - P(x))$
$= (0.4 - 0.8 \cdot 0.3)/(1 - 0.3) = 0.16/0.7 = 8/35 \approx 0.229$

To get the points, it was sufficient to sketch this procedure. A full calculation was not required.

3. The topic you are interested in was covered by a tutorial. Using Bayes' rule, calculate the exact probability that it will come up in the exam. 3 Points

Solution: $P(x \mid t) = P(t \mid x) \cdot P(x)/P(t) = 0.8 \cdot 0.3/0.4 = 0.6$

Problem 2.2 (Bayesian Network)
Consider the following Bayesian network where all random variables are Boolean:



We write X^+ and X^- for the events that X is true/false.

1. What is the *minimal* number of entries that you have to store to represent all conditional probability tables of this network? 2 Points

Solution: 8 (1 each for A and E, 4 for S, 2 for H) It would be 16 if we did not leverage redundancy.

2. Give $P(H^+ \mid A^+,E^+)$ in terms of the entries of the conditional probability tables of the network. 3 Points

Solution: $P(H^+ \mid A^+,E^+) = P(H^+ \mid S^+) \cdot P(S^+ \mid A^+,E^+) + P(H^+ \mid S^-) \cdot P(S^- \mid A^+,E^+)$

From now on assume that
- A and E may each independently cause S.
- S may cause H.
- There are no other causes of S.

3. In what ways (if at all) does this information on the causal relationships constrain the possible values of the entries of the conditional probability table of S? 2 Points

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Solution: $P(S^+ \mid A^-, E^-) = 0$ (no other causes). $P(S^+ \mid A^+, E^-)$ and $P(S^+ \mid A^-, E^+)$ are non-zero (possible causes). $P(H^+ \mid S^+)$ is non-zero (possible cause).

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4. Give an example of a situation (by giving an informal meaning for each random variable) that can be modeled by the above network and satisfies the above assumptions about causality and in which $P(S^+ | A^+, E^+) = 0$. 2 Points

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Solution: For example,
- H: There is a noise.
- S: An object falls over.
- A/E: The object is pushed equally from the left/right in such a way that both pushes cancel each other out.

The key property is that A and E may cause S but cancel each other out if they are both present.

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3 Markovian Reasoning

Problem 3.1 (Hidden Markov Models)
Consider the following situation:
- You make daily observations about your business B. Each day business is either good (b_1) or bad (b_2).
- You know this is caused by the general economic situation G, which you cannot easily observe, and which can be getting worse (g_1), be stable (g_2), or getting better (g_3).
- You have previously obtained the following information:
  - when the economy gets worse, your business is good 36% of the time,
  - when the economy is stable, your business is good 84% of the time,
  - when the economy gets better, your business is good 90% of the time,
  - half the time, the economy is the same as on the previous day,
  - when the economy changes from one day to the next, each change is equally likely.

You want to model this situation as a hidden Markov model with two families of random variables indexed by day number d.

1. Give the state and evidence variables and their domains. 2 Points

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Solution: State variables $G_d \in \{g_1, g_2, g_3\}$, evidence variables $B_d \in \{b_1, b_2\}$

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2. How can you tell that the sensor model is stationary here? 1 Points

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Solution: The business-economy relation is the same for each day.

3. Complete the following sentence: The transition model T is given by the matrix 2 Points

$$T = \left(\quad \right) \qquad \text{where} \qquad T_{ij} = P(G_{d+1} = g_j \mid G_d = g_i).$$

Solution: $T = \begin{pmatrix} 0.5 & 0.25 & 0.25 \\ 0.25 & 0.5 & 0.25 \\ 0.25 & 0.25 & 0.5 \end{pmatrix}$

4. Complete the following sentence: The sensor model S is given by the matrix 2 Points

$$S = \left(\quad \right) \qquad \text{where} \qquad S_{ij} = P(B_d = b_j \mid G_d = g_i).$$

Solution: $S = \begin{pmatrix} 0.36 & 0.64 \\ 0.84 & 0.16 \\ 0.9 & 0.1 \end{pmatrix}$

5. Let T be as above and let $\mathbf{v}$ be a 3-dimensional vector whose coefficients sum to 1. What is the intuitive meaning of the property $T \cdot \mathbf{v} = \mathbf{v}$? 2 Points

Solution: $\mathbf{v}$ is a probability distribution of the economy that is a fixed point of the transition model, i.e., the distribution will stay the same when predicting the future.

The economy got better yesterday, and your business is good today. You want to use filtering to obtain the probability distribution of today's economy.
You proceed as follows:

6. Give the recursive filtering equation for $f_{1:d+1}$. 1 Points

Solution: $f_{1:d+1} = \alpha(O_{d+1} \cdot T^{\mathrm{T}} f_{1:d})$ (where T^{T} is the transpose of the matrix T)

7. Give the initial value $f_{1:0}$ to use in this case. 1 Points

Solution: $f_{1:0} = \langle 0, 0, 1 \rangle$

8. Give the diagonal sensor matrix O_1 to use in this case. 1 Points

Solution: $O_1 = \begin{pmatrix} 0.36 & 0 & 0 \\ 0 & 0.84 & 0 \\ 0 & 0 & 0.9 \end{pmatrix}$

Problem 3.2
Consider the following agent in a stochastic environment. The agent moves on a cyclic track of 7 states (0 to 6). Its possible actions are:

- -1 (move counter-clockwise) or 1 (move clockwise), which make no movement if they fail
- 2 (move clockwise twice), which moves one step counter-clockwise if it fails

All actions succeed with probability 0.6 and fail otherwise. The agent starts at 3. Its goal is to reach 0. We use a reward of 10.0 for the goal and -2.0 otherwise, and a discount factor of $\gamma = 0.5$.

1. Give the sets S of states and A of actions for this agent, modeled as an MDP. 2 Points

   Solution: $S = \{0, 1, 2, 3, 4, 5, 6\}$, $A = \{1, -1, 2\}$

2. State the Bellman equation for $U(s)$ for $s \in S$. 2 Points

   Solution:
   $$U(s) = R(s) + \gamma \max_{a \in A} \sum_{s'} P(s' \mid s, a) U(s')$$

3. Assume current utilities $U(s) = 0$ for all states. Perform one value iteration step to calculate $U(4)$. 2 Points

   Solution: With $U = 0$, future rewards are 0. Thus:
   $$U(4) = R(4) = -2$$

4. Suppose we solved the utilities as $U^*(s)$. How do we derive the optimal action $\pi^*(s)$ for state s? 2 Points

   Solution: We calculate the utilities for all actions and pick the action that maximizes utility. Formally:
   $$\pi^*(s) = \text{argmax}_a \sum_{s'} P(s' \mid s, a) U^*(s')$$
   It was sufficient but not necessary to give the formula.

5. Now assume we use the corresponding POMDP with current belief state $b(3) = 0.5, b(0) = 0.5$, and we apply action $a = -1$. Calculate the new belief state b'. 2 Points

   Solution: $b'(2) = 0.3$, $b'(3) = 0.2$, $b'(6) = 0.3$, $b'(0) = 0.2$
   An alternative reading of the question leads to the solutions $b'(4) = 0.3$, $b'(3) = 0.2$, $b'(1) = 0.3$, $b'(0) = 0.2$, which was also accepted. Additionally, $b'(0) = 0.5$ (no movement after reaching goal) is not technically correct but was accepted.

4 Learning

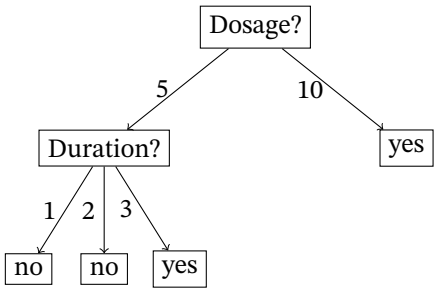
Problem 4.1 (Learning for Data with Attributes)
Consider the following data observed for some patients during a medical trial of a new drug:

Duration of Treatment 1, 2, or 3 weeks	Dosage 5 or 10 mg/week	Success yes/no
1	5	no
1	10	yes
2	5	no
2	10	yes
3	5	yes
3	10	yes

1. Give the decision tree for the attribute order "Dosage, Duration" that determines the treatment's success given the duration and dosage of the treatment. 3 Points

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Solution:



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2. Give the entropy of the attribute "Success". 1 Points

Solution: $-1/3\log_2 1/3 - 2/3\log_2 2/3 = \log_2 3 - 2/3$

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3. Give the entropy of the Boolean attribute that the dosage was a multiple of 5. 1 Points

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Solution: 0

To give a logical formulation, we use binary predicate symbols *Dur* and *Dos* and a unary predicate symbol *Suc*. For example, $Dur(p,2) \wedge Suc(p)$ represents that patient p was successfully treated for 2 weeks.

4. Give a formula in first-order logic that represents the rule that treatments with dosage 5 were not successful. 2 Points

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Solution: $\forall p.Dos(p,5) \Rightarrow \neg Suc(p)$
Using equivalence instead of implication was also accepted.

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5. Now assume we have learned the rule $\forall p.Suc(p) \Leftrightarrow (Dur(p,2) \vee Dur(p,3))$. What is its $0/1$ loss? 2 Points

Solution: 2 (The formula is falsely negative once and falsely positive once.)

Problem 4.2 (Inductive Learning through Linear Regression)
Consider a list of examples $(\vec{x}_i, y_i) \in \mathbb{R}^n \times \mathbb{R}$ for $i = 1, \ldots, m$.

1. What is the hypothesis space for applying linear regression here? 2 Points

Solution: The set of linear functions $\vec{x} \mapsto \vec{w} \cdot \vec{x} + w_0$. It was not necessary to give the formula with the parameters; in that the type of the functions must be given clearly.

2. Assuming m is large, briefly explain under what circumstances we can or cannot find a consistent hypothesis. 2 Points

Solution: If the examples are all sampled from a fixed linear function without measurement error, there is exactly 1 consistent hypothesis (and it is easy to find). If the target function is not linear or the examples have some error, there is no consistent hypothesis at all.

3. Regarding loss functions, briefly explain the (dis)advantages of using error rate and squared error loss in this setting. 2 Points

Solution: $L_{0/1}$ (error rate) is not a good choice. It just counts how often $h(\vec{x}_i) \neq y_i$, but equality comparisons on real number data are typically not reliable.
Instead, we should a loss function that measures how far off a hypothesis is. $(h(\vec{x}_i) - y_i)^2$ (squared error) is a typical choice.

4. Now assume $m = 2$. Give the resulting formula for the squared error loss in this setting. 2 Points

Solution: $1/2 \cdot ((\vec{w} \cdot \vec{x}_1 + w_0 - y_1)^2 + (\vec{w} \cdot \vec{x}_2 + w_0 - y_2)^2)$

5. Briefly explain why we are interested in taking the derivative of the loss with respect to the regression parameters. 1 Points

Solution: It allows solving for the values of the parameters, the w_i above, for which the loss is minimal.

Problem 4.3 (Reinforcement Learning)

1. Consider a fully observable Markov Decision Process and a fixed policy π. Briefly explain the similarities and differences between policy evaluation and passive adaptive dynamic programming (passive ADP). 2 Points

Solution: Both aim to find the utility of a fixed policy π by using the Bellman equation. But in ADP, the agent does not know the transition model and the reward function. Instead, they are obtained through its percepts.

Now we apply passive ADP. Assume the states are $\{0, \dots, 3\}$, and π always picks the same action a. Our agent performs one trial moving according to π. It observes the states

$$0, 0, 1, 1, 1, 2, 2, 1, 2, 2, 3, 3$$

2. What transition model T does the agent learn from these observations? Give T as a list of distributions $T(s, a)$ for each s. 3 Points

Solution:

$$\begin{aligned} T(0, a) &= [0.5, 0.5, 0, 0] \\ T(1, a) &= [0, 0.5, 0.5, 0] \\ T(2, a) &= [0, 0.25, 0.5, 0.25] \\ T(3, a) &= [0, 0, 0, 1] \end{aligned}$$

3. Now assume the agent additionally observes the reward after each action during the above trial. After the last 2 actions, it observes the reward 1.0; after all other actions, it observes the reward -0.1. 2 Points
What reward function does it learn from this?

Solution: $R(3) = 1.0$ and $R(s) = -0.1$ otherwise.

4. Briefly explain the main change when generalizing to *active* ADP? 1 Points

Solution: There is no fixed policy. We have to try all actions to learn the transition model and eventually use that to find the optimal policy.

5 Natural Language Processing

Problem 5.1 (Language Models)
Consider a language L over the alphabet with characters a, b, c, d, and consider a corpus D for it consisting of the five words

$$a\,b\,c\,c\,c \qquad a\,b\,b\,d\,a \qquad a\,a\,a\,b\,c \qquad b\,c\,d\,b \qquad a\,a\,b\,a\,c$$

You want to build a 3-gram model for L.

1. How many 3-grams are there over L? 1 Points

Solution: $4^3 = 64$

2. Now assume you store the 3-gram model as a Bayesian network with variables w_1, w_2, and w_3 (representing the 3-gram $w_1\, w_2\, w_3$), how many entries does the conditional probability table for w_3 have? 2 Points

Solution: $(4 \cdot 4) \cdot 4 = 64$ or $(4 \cdot 4) \cdot 3 = 48$ if we utilize redundancy. Either answer was accepted.

3. Give the numerical value of $P(w_3 = a \mid w_{1:2} = aa)$ in the resulting model. 2 Points

Solution: $1/3$

4. Give the numerical value of $\texttt{tf}(a, cdad)$. 1 Points

Solution: $1/4$

5. Give the numerical value of $\texttt{idf}(a, D)$. 1 Points

Solution: $\log_{10} 5/4$

6. Now you want to use the `tfidf` vector embedding. What are the objects that are embedded, and how are the vectors computed using `tf` and `idf`? 2 Points

Solution: We embed documents/words W from L. The vector for W is $\texttt{tf}(x, W) \cdot \texttt{idf}(x, D)$ for $x \in \{a, b, c, d\}$.

7. Now assume you are interested in all documents that contain the subword $b\, c$. A query system returns all 5 documents. 2 Points
   Give the numerical values of
   1. precision: `$3/5$`
   2. recall: `$1$`

Solution: filled in above